Fakultät Informatik

From CAD Software to Slicer

Workflow Optimization using Print Hints in 3MF Files for 3D Printing

Bachelorarbeit im Studiengang Medieninformatik

vorgelegt von

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Abstract

Kurzdarstellung

//TODO

Contents

Chapter 1.

Introduction

> It is one of the clearest paradoxes of modern work that computer-based systems designed to reduce task complexity and cognitive workload actually often impose even greater demands and stresses on the very individuals they are supposed to be helping.[1, p. 222]

When designing a model for 3D printing, the process frequently follows a certain set of iterative steps. After the ideation phase, the design engineer first creates the model using CAD software, imports the file (e. g. STL, 3MF) to a slicer software, sets the necessary parameters and finally starts printing. They evaluate the result for design flaws or areas of improvement, leading to model revisions and thereby reinitiating the workflow as pictured in figure 2.1. Throughout the iterations, some printing parameters will be tweaked but overall you want to keep them consistent to have a reliable comparison. Despite automation and optimization having been a prevalent topic in the software developer community for years, setting the parameters is generally a manual step. Hence, the design engineer has to remember their settings or save them some other way, possibly by writing them down meticulously or taking screenshots.

With access to printers in public institutions becoming easier and the growing popularity of 3D printing not only within the *maker* community but in casual users as well, HCI needs to focus on streamlining error-prone workflows like the one described above[2]. "[C]urrent tools that focus on supporting only one aspect of the workflow (*e. g.* , only modelling or only printing) may be less useful for casual makers and there is opportunity in HCI research to innovate on design tools that take into account the interdependencies within the 3D printing workflow"[2]

This thesis focuses on the communication between CAD software and slicer, with a particular focus on proving the feasibility of 'print hints'. The goal is to streamline the iterative design workflow by reducing the need to repeatedly configure the same printing parameters in each iteration, therefore lowering cognitive workload for design engineers. The approach as currently planned contains two core components: firstly, an improved export function within the CAD software *Autodesk Fusion*, e. g. to mark certain objects

as modifiers; and second, automatic handling of materials and these print hints within the open source slicer software called *OrcaSlicer*.

basic (HCI) aspects in which I want to improve the workflow (possibly to be referenced in the following texts) → what does it solve

- sharing files with slicer but not printer specific settings
- how it reduces cognitive workload in repetitive steps

"*makers*, a community of enthusiasts who focus on fabrication, invention and experience sharing, and who collaborate and learn in environments known as *makerspaces*"[2]

"thousands of public institutions, such as schools, libraries, and universities are now setting up print centers or creative hubs for end users to make use of digital fabrication technologies"[2]

"makerspaces that are often run by enthusiasts [...] attract engineers, entrepreneurs, inventors, hackers, craftsmen, and artists"[2]

"casual makers serve as a proxy for what it will be like for the general public to use 3D printing"[2]

"if HCI is going to be at the forefront of inventing new fabrication design tools and interfaces[...], we need to look beyong enthusiasts and understand the challenges and opportunities that exist in facilitating the interactions of casual makers with fabrication technology"[2]

"current tools that focus on supporting only one aspect of the workflow (*e. g.* , only modelling or only printing) may be less useful for casual makers and there is opportunity in HCI research to innovate on design tools that take into account the interdependencies within the 3D printing workflow"[2]

"while lowering the usability barriers of fabrication tools is important, it is perhaps even more important to weave-in expertise on best practices and troubleshooting tips within the walk-up-and-use tools used by casual makers to reduce theiir dependecy on operators"[2]

"opportunities that exist in building the next generation of inter-connected fabrication tools with features for supporting expertise sharing"[2]

"enthusiasts and professional users—that is, they were trained or technically literate in fabrication"[2]

Chapter 2.

Problem

what's the real world problem I am trying to solve? why is solving it important?

- Repetitive Steps in the Iterative Workflow

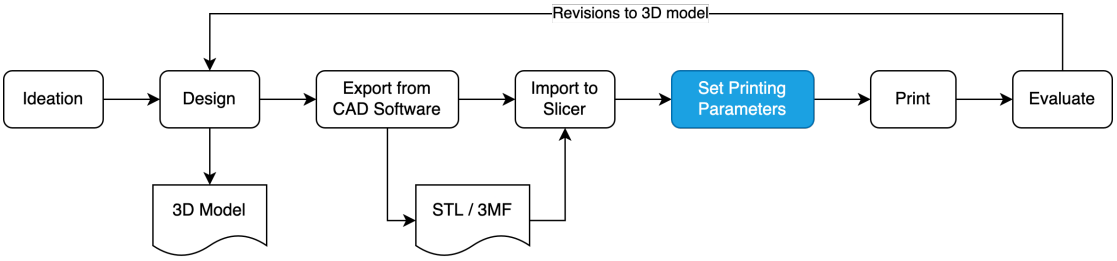

Figure 2.1.: Iterative workflow in 3D printing highlighting the step to be optimized

- Negative Effects resulting from manual Repetitive Steps

  //TODO research cognitive workload / increased amount of errors due to necessary repetitive steps (podcast about to-do lists → air plane take off / landing checklists (yes, this is relevant))

  some HCI foundational work if necessary

  "If the user's print failed or did not match their design intent, they would have to adjust either their model or print settings to correct the issue, often with the operator's assistance"[2]

- Usage of Presets

  do people (esp. beginners) use presets? if no, why and how would it make their life easier? is it even possible in community spaces like libraries and schools?

- the need for elaborate and extensive printing instructions when sharing a complex model online

Chapter 3.

Background

3.1. State of the Art

- current situation → workflow as is with specific, tangible examples; how do others solve this?
- related work if applicable

When analyzing the process from idea to print, several main steps can be identified. Using the simple box in 3.1 with some particular features as an example, the following analysis will focus on the steps from exporting a model to printing, while taking the ideation and design phase as given.

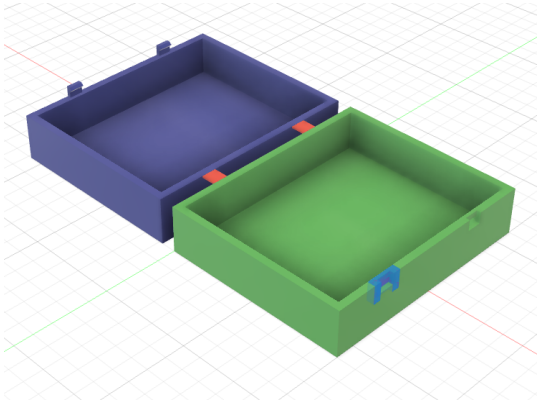

Figure 3.1.: 3D model of simple box

3.1.1. Workflow using STL

When exporting the model as an STL file,

3.1.2. Workflow using 3MF

3.1.3. Workflow using OBJ

Once a 3D model has been created, the user exports it using one of several ways Fusion offers. Depending on the file format, a model can be exported as one file containing multiple objects or one file per object. The three formats used most often for 3D printing are STL, 3MF and OBJ, each presenting different advantages and disadvantages. The user then opens the exported file in OrcaSlicer. Opening the file, or importing it, may

cause different dialogs to open, depending on the file type and content. So far all steps are fairly simple.

Next, various printing parameters have to be applied. Especially when a model consists of multiple parts, the parameters might vary from one object to the next. Firstly, materials can be set using system or user defined presets per object. Material settings entail for example the filament itself, nozzle temperature, general printing speed and cooling settings. OrcaSlicer actively asks the user, whether they want to save any changes to material parameters they have made, which makes it easy to save either parameters specific to one's printer and filament or even specific to one project while simultaneously keeping default material presets untouched.

Furthermore settings like the amount of wall loops, infill percentage and pattern, and printing speed for single layers might need to be customized. It is possible to save these as presets as well, however only globally, i. e. they can only be applied to the entire project, not to specific objects within. As criticized by GitHub user *headamame* in the [6, OrcaSlicer Issue #7769], particularly for projects containing multiple objects, using presets is not a valid option.

//TODO different workflows for different file formats

1. create 3D model in Fusion
2. export as 3mf file
3. open in OrcaSlicer
4. set parameters
    - apply system or user defined material presets, optionally setting up new presets
    - apply settings (wall loops, infill percentage and pattern, layer speed etc.) manually per object or using presets globally (only available globally, not for single objects → OrcaSlicer Issue #7769)
    - change type to Modifier / Support Blocker / Support Enhancer; in case of Modifier → apply settings accordingly

3.2. Foundations

- what is CAD software?

- what is a slicer?

  "Processing the 3D model occurred in specialized software known as a slicer (as it slices the model into layers for the printer), which allowed users to select print settings, such as layer height and print speed"[2].

- 3MF File Format → why 3MF and not STL or other file formats? why not OBJ, since it supports exporting materials from fusion and mapping then in OrcaSlicer natively?

3.2.0.1. The 3MF specification

[3]

- "3MF Document markup has been designed in anticipation of the evolution of this specification. It also allows third parties to extend the markup. Extensions are a critical part of 3MF, and as such, this core specification is as narrow as possible." [p. 8]
- 3MF = ZIP archive following the .ZIP File Format Specifications by PKWARE Inc. containing one or more 3D payloads
- 3D payload = "complete collection of interdependent parts and relationships within a package" consisting of "a 3D object definition and its supporting files"
- 3MF document parts → table 3.1, figure 3.2

```
example.3mf
├── _rels/
├── [Content_Types].xml/
├── 3D/
│   └── 3dmodel.model
└── Metadata/
    └── thumbnail.png
```

Figure 3.2.: Typical internal structure of 3MF file as exported from *Autodesk Fusion*

- PrintTicket → seems to be OS specific, so probably not helpful but //TODO investigate further
- most important: 3D Model Part = root of 3D payload → "contains definitions of one or more objects to be fabricated by 3D manufacturing processes"[3, p. 6] → contains resource definitions (objects, materials) and items to be build
  - there MUST be exactly one <model> element in a 3D Model part

Name	**Description**	**Required / Optional**
3D Model	Contains the description of one or more 3D objects for manufacturing.	Required
Core Properties	The OPC part that contains various document properties.	Optional
Object Thumbnail	Contains a small JPEG or PNG image that represents a 3D object in a 3D Model.	Optional
3D Texture	Contains a texture used to apply color to a 3D object in the 3D Model part (available for extensions)	Optional
PrintTicket	Provides settings to be used when outputting the 3D object(s) in the 3D Model part.	Optional
Custom Parts	OPC parts that are associated with metadata	Optional

Table 3.1.: Excerpt of 3MF document parts[3, p. 4]

- a model must have two additional child elements: <resources> and <build>
- <resources> element

  provides a set of definitions that can be drawn from to define a 3D object; root element of a library of constituent pieces of the overall 3D object definition → objects, properties, materials; 'object resource': 3d object that could be but will not be manufactured on its own

- <build> element

  provides a set of items that should actually be manufactured as part of the job; contains `<item>` elements = objects to be manufactured

Chapter 4.

Method

what's the solution(s) I came up with → sketching out different possibilities and why I chose the one I did

4.1. Explored Possible Solutions

4.2. Fusion Add-In

- what is CAD software
- a lil about their API: add-in vs. script, Python vs. C++
- why I chose to realise it as a Python add-in
- possibly different approaches if more than one way to solve it came up

As mentioned before, the software supports developers to create additional functionalities by providing an extensive API and the option to create either an add-in or a script directly from within Fusion's UI. Using this, Fusion sets up the project structure as well as some example code to make it easier to get started especially for beginners. Fusion runs natively on Python but offers to create the add-in or script either in Python or in C++.

4.2.1. Script vs. Add-in

[**?**]

- **Run on Startup** – This setting is add-in-specific and indicates if the add-in should be run automatically when Fusion is started. Most add-ins will want to take advantage of this capability so the commands they define will be available to the user as soon as Fusion starts.

- The user executes a script through the "Scripts and Add-Ins" command, and the execution stops immediately after the run function is complete. A script runs, and then it is done.
- An add-in is typically automatically loaded by Fusion when Fusion starts up. An add-in also usually creates one or more custom commands and adds them to the user interface during startup. The add-in continues to run throughout the Fusion session, so it can react whenever the user executes any of its commands. The add-in remains running until Fusion is shut down or the user explicitly stops it through the "Scripts and Add-Ins" dialog. When it stops, it cleans up whatever user interface customization it created in its stop function.

4.2.2. Python vs. C++

4.3. if necessary: changes to 3MF File Format

4.4. slicer changes

- how does it process 3mf per default?
- different possible approaches to solve the problem at hand and why did I do it the chosen way?

4.5. Implementation Design

- software architecture (diagram?)
- input, output
- choices that had to be made

Chapter 5.

Results

- what does the optimised workflow look like?
- why is the solution good?
  - how are the results analysed / validated?
  - openly available → advantages of that
  - but: it is a proof of concept / prototype and thus will most likely not be the end of the story

Chapter 6.

Conclusion

6.1. Outlook

- communication with OrcaSlicer's owner softfever?
- option of code being pulled into other slicers
- what is left to do?

6.2. Summary

Appendix A.

Supplemental Information

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <model xmlns="http://schemas.microsoft.com/3dmanufacturing/core
       /2015/02" unit="millimeter" xml:lang="en-US" xmlns:m="http:
       //schemas.microsoft.com/3dmanufacturing/material/2015/02"
       xmlns:p="http://schemas.microsoft.com/3dmanufacturing/
       production/2015/06" xmlns:b="http://schemas.microsoft.com/3
       dmanufacturing/beamlattice/2017/02" xmlns:s="http://schemas.
       microsoft.com/3dmanufacturing/slice/2015/07" xmlns:sc="http:
       //schemas.microsoft.com/3dmanufacturing/securecontent
       /2019/04">
3    <metadata name="Title" type="QName" preserve="1">Test v2</
         metadata>
4    <resources>
5     <m:colorgroup id="2">
6        <m:color color="#CC60FFFF"/>
7       <m:color color="#A0A0A0FF"/>
8      </m:colorgroup>
9     <object id="1" name="Body9" type="model" p:UUID="33f607be
         -287f-48ea-b893-8ee1626ed2b1" pid="2" pindex="0">
10      <mesh>
11       <vertices>
12         <vertex x="-39.484996" y="6.249999" z="30.000000" />
13         <vertex x="-37.568216" y="6.249999" z="30.000000" />
14         <vertex x="-37.568216" y="6.249999" z="33.000000" />
15         ...
16         </vertices>
17         <triangles>
18           <triangle v1="0" v2="1" v3="3" />
19           <triangle v1="3" v2="1" v3="2" />
20           <triangle v1="4" v2="0" v3="5" />
```

```
21              ...
22            </triangles>
23          </mesh>
24        </object>
25    </resources>
26    <build p:UUID="cd9553e5-fa90-4b4e-916e-bd1fed06979c">
27      <item objectid="1" p:UUID="91670927-ae59-46f3-cb97-91
            a310462033"/>
28    </build>
29  </model>
```

Listing A.1: Abbreviated Example of 3MF's 3dmodel.model File

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